Theory Lectures on

Manufacturing Processes (MEI102)

PAM – Plasma Arc Machining

(Lecture 8A)

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Content prepared for theory lecture for 1st Year BTech level students for common course on Manufacturing Processes.
Lecture duration ≈ 20 min

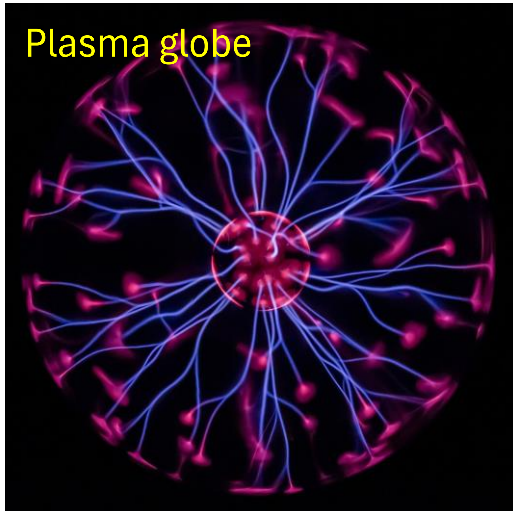
Classifications of NTM

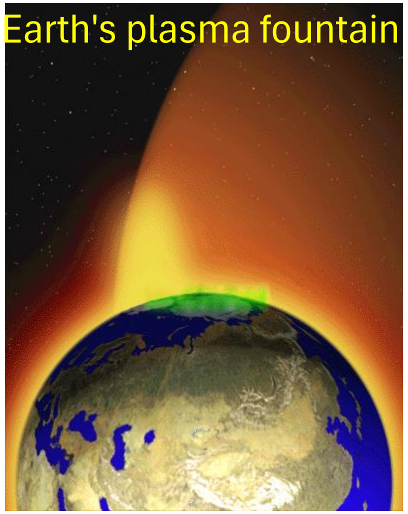
Non-traditional machining processes

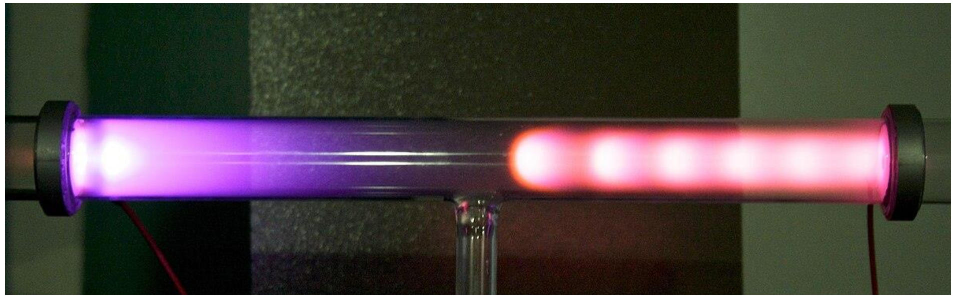
- **Mechanical**
  - Abrasive jet machining (AJM)
  - Ultrasonic machining (USM)
  - Water jet machining (WJM)
  - Abrasive flow machining (AFM)
  - Magnetic abrasive finishing (MAF)
- **Thermal**
  - Electrical discharge machining (EDM)
  - Wire electrical discharge machining (WEDM)
  - Electron beam machining (EBM)
  - Laser beam machining (LBM)
  - Ion beam machining (IBM)
  - Plasma Arc Machining (PAM)
- **Chemical and Electrochemical**
  - Chemical machining (CHM)
  - Electrochemical machining (ECM)
  - Photo chemical machining (PCM)
  - Biochemical machining (BM)
- **Hybrid**
  - Electrochemical discharge machining (ECDM)
  - Abrasive water jet machining (AWJM)
  - Electrochemical grinding (ECG)
  - Laser-assisted jet electrochemical machining (LA-JECM)

What is the plasma?

1. Plasma is called the fourth state of matter after solid, liquid, and gas.
2. Consider a mass of any solid matter – it will consists of a number of atoms – although very high but finite number of atoms. If the temperature of this ***solid*** matter is increased, then inter-atomic space between the atoms will increase resulting in volumetric expansion. Once the temperature exceeds melting temperature, the same matter changes its phase to ***liquid*** that has higher volume than solid counterpart. The number of atoms also remains unchanged. If temperature is further increased, the same liquid will change its phase to ***gas*** without any change in the number of atoms. Further increase in temperature will result in dissociation of electrons from atoms (ionization) that develops a new phase (***plasma***) without any change in the total number of atoms.
3. A plasma is a mixture of positively and negatively charged particles, ions and electrons, in which the negatively charged electrons are almost completely neutralized by the positively charged ions. Thus the whole plasma is electrically neutral. Plasma contains neutrally charged particles as well.
4. A plasma is a quasi-neutral gas because the overall electrical charge of a plasma is roughly zero but free charge-induced electromagnetic forces exist. Moving charged particles generate electric currents, and any movement of a charged plasma particle affects and is affected by the fields created by the other charges.
5. As the negative and positive charges are free to move, plasma is an excellent conductor and is influenced by magnetic and electric fields.







Vacuum and Mean Free Path

Vacuum range	Pressure in mmHg (1 mmHg = 760 atm)	Number density (Molecules / m^3)	Mean free path
Ambient pressure	759.8	2.7×10^{25}	64 – 68 nm
Low vacuum	$220 – 8\times10^{-1}$	$10^{25} – 10^{22}$	0.1 – 100 μm
Medium vacuum	$8\times10^{-1} – 8\times10^{-4}$	$10^{22} – 10^{19}$	0.1 – 100 mm
High vacuum	$8\times10^{-4} – 8\times10^{-8}$	$10^{19} – 10^{15}$	10 cm – 1 km
Ultra-high vacuum	$8\times10^{-8} – 8\times10^{-13}$	$10^{15} – 10^{10}$	1 km – 10^5 km
Extremely high vacuum	$<8\times10^{-13}$	$<10^{10}$	$>10^5$ km

A **vacuum** is space devoid of matter.

- It is never possible to get absolute vacuum.
- A typical vacuum cleaner produces enough suction to reduce air pressure by around 20% of standard atmospheric pressure.
- Ultra-high vacuum chambers operate below one trillionth (10^{-12}) of atmospheric pressure.

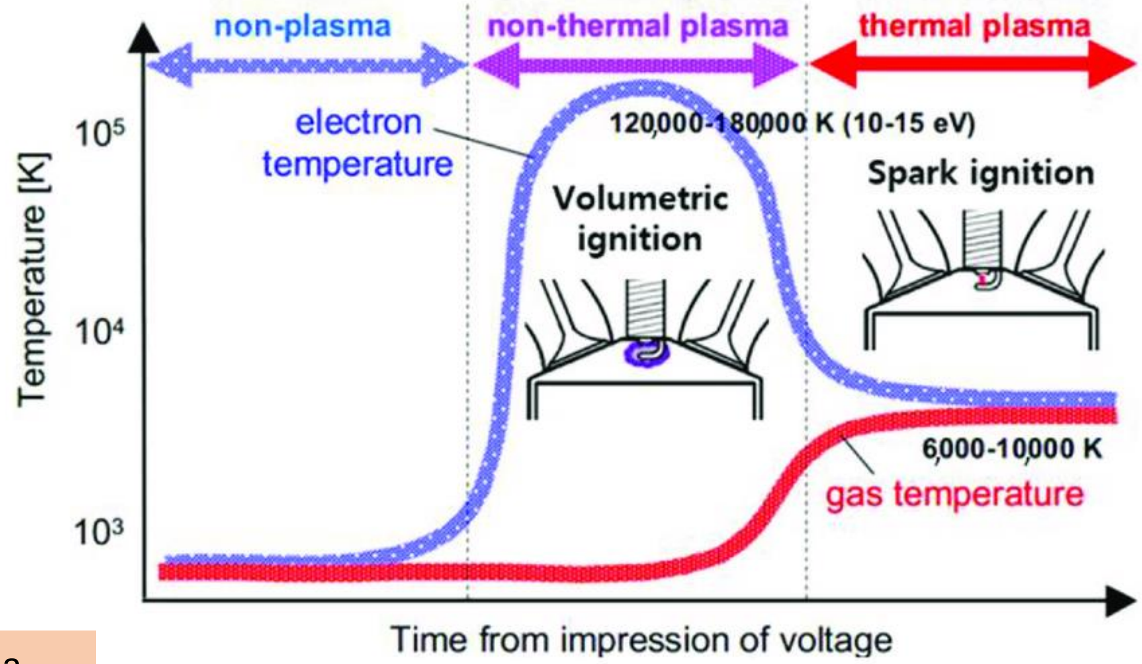
Mean free path is the average distance over which a moving particle (such as an atom, a molecule, or a photon) travels before substantially changing its direction or energy as a result of one or more successive collisions with other particles.

- Vacuum (on engineering applications) is just one perception of low pressure (that results in low atom density).
- State of the matter can be altered by changing pressure without changing temperature (example: CNC gas cylinder where gaseous substance is pressurized to remain in liquid state in normal room temperature). When pressure is increased on a matter, its interatomic distance reduces and thus atom number density (number of atoms per unit volume) increases.
- Getting absolute vacuum on Earth's surface is practically challenging.

Plasmas of technological interest can be broadly divided into two categories

Glow discharge plasma (non-thermal plasma)

- It is low pressure plasma characterized by high electron temperatures and low ion and neutral temperatures.
- Commonly used in household lights, surface cleaning, etching, film deposition, controlling chemical reactivity, etc.
- Electric current flows through the tube in a low-pressure arc discharge. Electrons collide and ionize noble gas atoms inside the bulb surrounding the filament to form a plasma. As a result of avalanche ionization, the conductivity of the ionized gas rapidly rises, allowing higher currents to flow through the lamp.



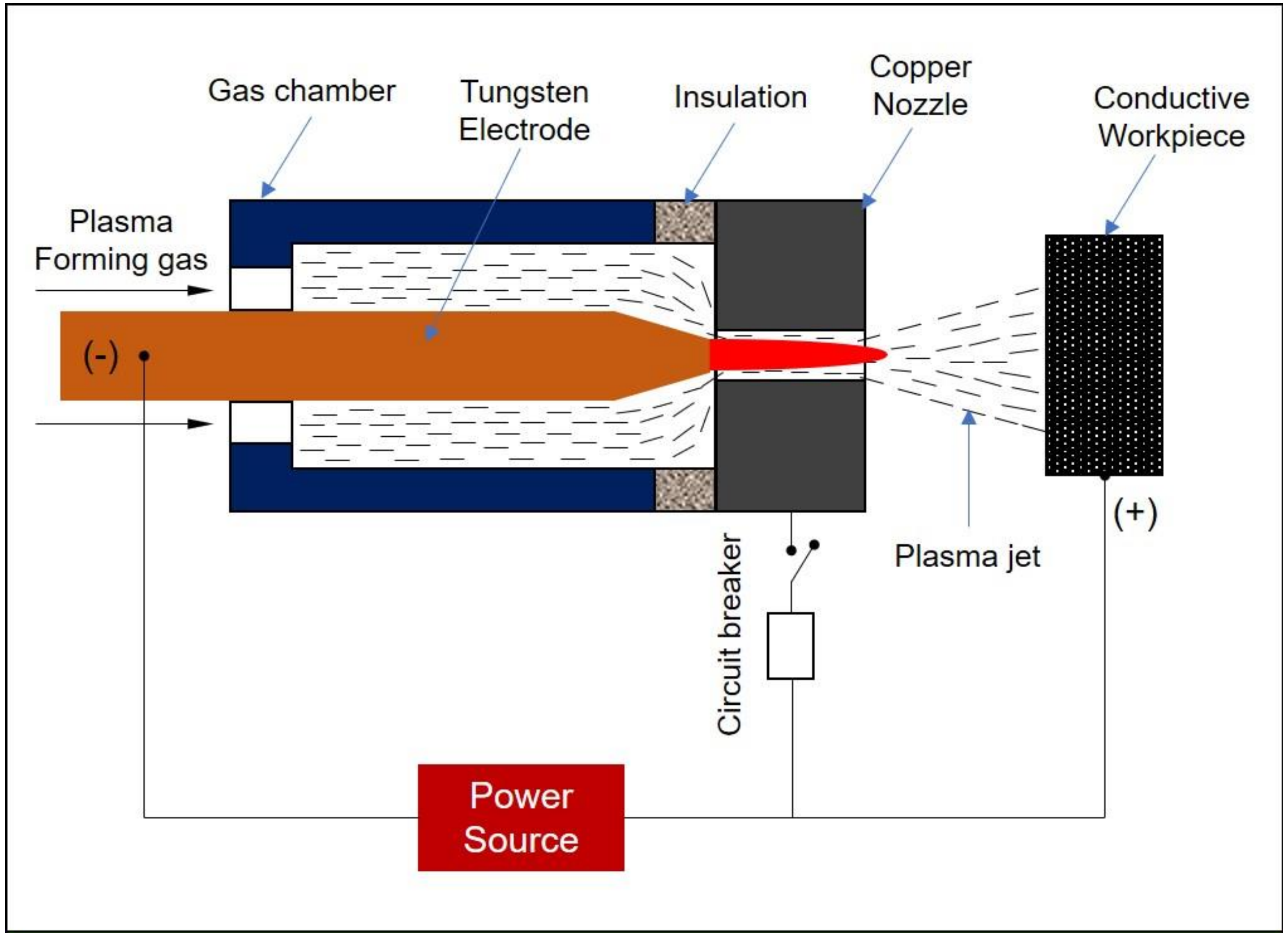
Fluorescent lamps – Glow discharge

Plasma spraying – Thermal plasma

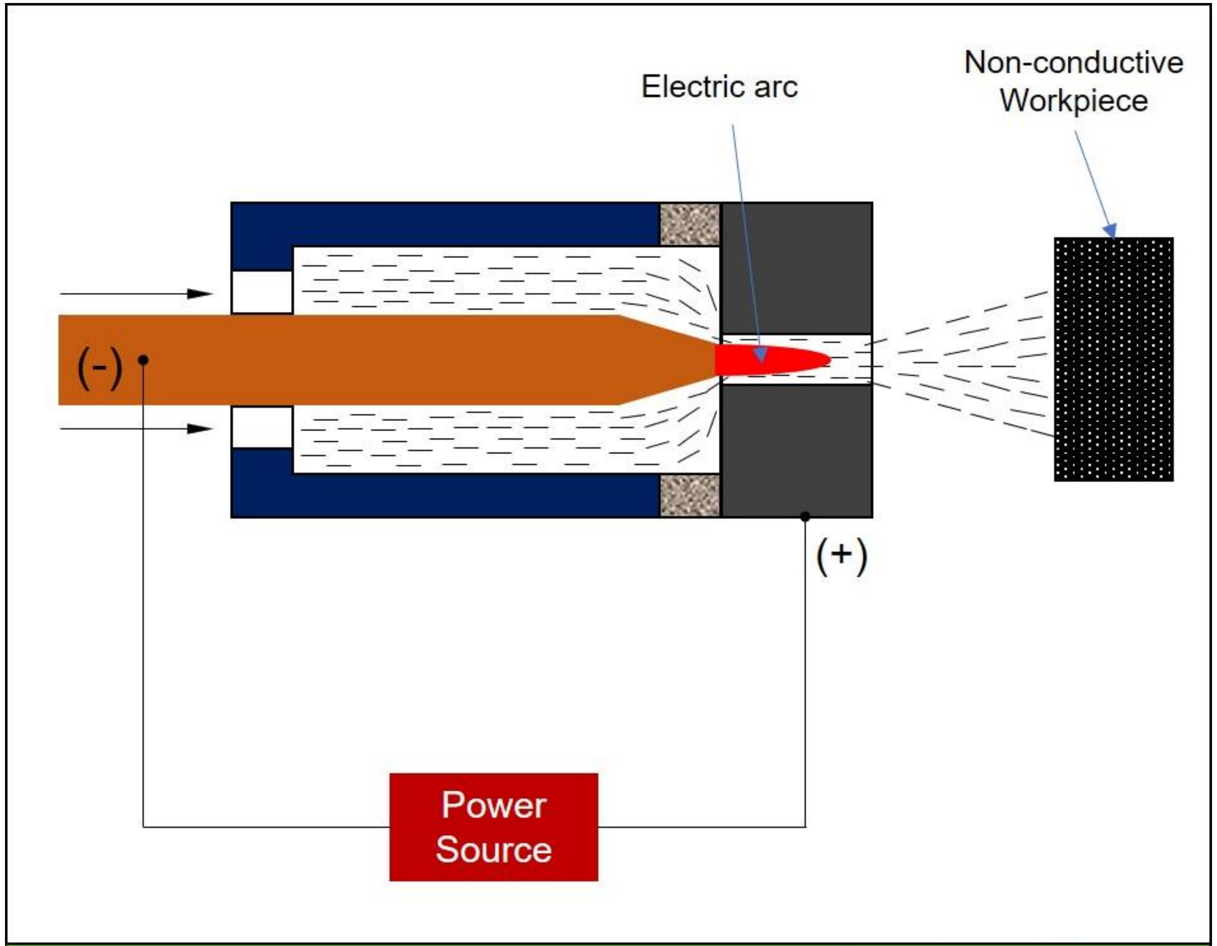
Thermal plasma (used in PAM)

- Thermal plasmas are atmospheric pressure plasmas characterized by high enthalpy content and temperatures around 2,000 – 20,000 °C.
- The main constituents (ions, electrons, neutrals) of a thermal plasma are in thermodynamic equilibrium and can be characterized by a single temperature.
- Thermal plasmas are widely used in cutting, welding and spraying (plasma coating).

Transferred Arc and Non-Transferred Arc in plasma formation



Transferred arc plasma torch
Direct arc plasma torch



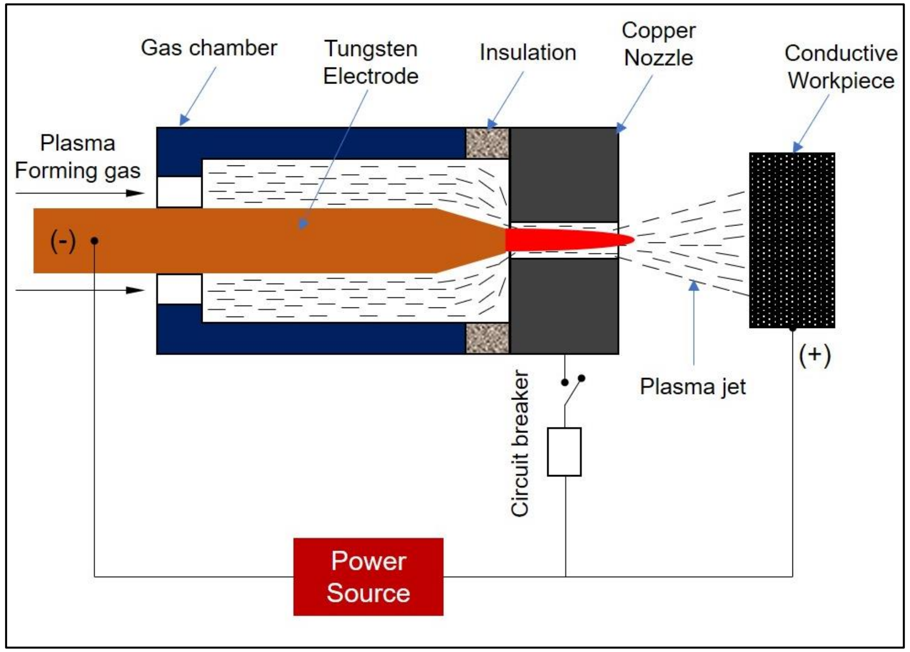
Non-transferred arc plasma torch
Indirect arc plasma torch

- For Plasma Arc Machining (PAM), thermal plasma is generated by electric arc heating of a suitable “plasma forming gas”. Thus presence of an electric arc is mandatory.
- This arc can be established between a tungsten electrode and the workpiece if it is electrically conductive (called *Direct arc*). If workpiece is non-conductive then the arc is established between the tungsten electrode and the copper nozzle (*Indirect arc*).

Difference between Transferred Arc and Non-Transferred Arc

Transferred Arc Plasma Torch	Non-transferred Arc Plasma Torch
The electric arc is constituted between an electrode and the workpiece. However, an auxiliary arc is established between the electrode and nozzle at the beginning of the work for a very short period.	The electric arc is constituted between an electrode and the nozzle, and the same arc is continued for the entire operation.
Here the workpiece is made anode (positive terminal of DC power source), whereas the nozzle is kept electrically neutral. Cathode is always a copper electrode.	Here the workpiece is kept electrically neutral, whereas nozzle is made anode. As usual, cathode is always a copper electrode.
Direct arc plasma torch can be applied to electrically conductive workpieces only.	Indirect arc plasma torch can be applied to every workpiece regardless of electrical conductivity. However, it is preferred for non-conducting materials.
Direct arc has relatively higher electro-thermal efficiency (85 – 95%).	Indirect arc has comparatively low electro-thermal efficiency (65 – 75%).
Direct arc is overwhelmingly used for machining (or cutting), welding, hardfacing, remelting, and spraying.	Indirect arc is preferred for flame spraying, spheroidizing heat treatment, ore processing, etc.
Transferred arc plasma torch is also known as “Direct Arc Plasma Torch” because the arc is maintained directly between the electrode and workpiece.	Non-transferred arc plasma torch is also known as “Indirect Arc Plasma Torch” because the arc is not maintained between the electrode and workpiece though the workpiece receives the heat.

Working principle of Plasma Arc Machining (PAM) or Plasma Arc Cutting (PAC)



Working principle of Plasma Arc Machining (PAM)

- A water-cooled copper nozzle is placed between the tungsten electrode and workpiece. This nozzle constricts the arc into a small cross-section that increases both the arc temperature and voltage.
- A suitable plasma-forming gas is forced to flow surrounding the high-temperature arc. As a result, this gas gets ionized resulting in the formation of high-temperature plasma jet. As the gas flow rate remains high, the plasma jet also has high velocity.
- This constricted high-velocity plasma jet that come out of the nozzle is directed towards the workpiece for necessary machining or cutting action.
- The plasma jet temperature remains very high of the order of 10,000 °C. Due to intense temperature, the plasma jet can instantly melt and vaporize the workpiece.
- Thus PAM is one thermal energy based non-traditional machining.
- The mechanism of material removal is thermal melting. Accordingly, material gets removed in the form of molten metal (assisted by plasma jet blow).
- This high velocity plasma jet blow away the molten metal from the cutting region. Accordingly, tendency of re-deposition and re-solidification of molten metal is low (unlike EDM where molten metal redeposit on work surface). As molten metal can be quickly blown away by the plasma jet, the material removal rate (MRR) remains reasonably high. Such blowing of molten metal also helps cutting thick metallic samples.
- The PAM process is apparently similar to that of oxy-fuel cutting widely used in civil areas. However, oxy-fuel cutting utilizes the heat from chemical reaction (oxidation), whereas, PAM utilizes heat energy of plasma itself.

Electrode material and plasma forming gas

Primary materials used for PAM electrodes

Copper

- Properties: Excellent electrical conductivity and heat dissipation.
- Use: Forms the main body of the electrode, providing a base for high-durability inserts.

Hafnium

- Properties: High melting point, excellent durability, and conductivity.
- Use: Often used as an insert at the electrode tip to enhance performance and lifespan.

Tungsten

- Properties: High melting point and good conductivity.
- Use: Used in some electrodes for specific applications for its durability under high temperatures.

Plasma forming gas (the gas that changes its phase from gaseous to plasma)

- Hydrogen – high heat capacity and high heat conductivity offer higher cutting speed. But risk of explosion and costly.
- Ammonia – explosion proof, cheaper.
- Hydrogen and argon mixture – Helps insulating the nozzle walls and protect it from overheating.
- Air – Cheaper but risk of oxidation.

Advantages, disadvantages, and applications

Advantages of PAM

- PAM process can be applied for cutting electrically conducting as well as non-conducting materials.
- PAM can be used for thicker materials.
- Underwater cutting is also possible by PAM process. There are certain advantages of under water PAM such as less noise, less fumes, high solubility of removed materials with water and less effect of oxidation by selecting proper plasma forming gas with respect to workpiece materials.
- Thick samples can be cut efficiently.
- No harmful chemical is used.

Applications of PAM

- Thick cutting of metal and non-metals.
- Underwater cutting.
- Plasma heating before conventional machining.

Disadvantages of PAM

- Rough surface finish – not suitable for finishing, rather suitable for rough cutting.
- High heat can induce thermal damage on workpiece.
- The large power supplies is required to cut thicker materials at high speed.
- Chance of re-deposition of metal on machined edges.
- Dimensional accuracy of cutting is less and difficult to generate sharp corners.
- Not suitable for cutting heat-sensitive materials such as magnesium, lead or zinc alloys.
- PAM emits strong light that can cause damage to human eyes. Operators must wear appropriate goggles or helmets with protective filters to protect their eyes during operation.

PAM videos

- https://www.youtube.com/shorts/rvJhKyugX_U?feature=share
- https://www.youtube.com/shorts/uii2mgyhcHw?feature=share
- https://www.youtube.com/shorts/8-oFqfRkEL8?feature=share
- https://www.youtube.com/watch?v=8i_EAm6Ob3A
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- https://www.youtube.com/watch?v=0CqcwH1Xjoo